

SIMATS ENGINEERING
ASSESSMENT TOOL 2 (LOW) - Industry Problem Solving Task

COURSE NAME: R PROGRAMMING

COURSE CODE: ITA04

COURSE OUTCOME COVERED

CO2: R Programming Structures, Different types of loops such as: for (), while() loops, Control Statements, Looping Over Non-vector Sets, - If-Else, Arithmetic and Boolean Operators and values, Default Values for Argument, Return Values, Deciding Whether to explicitly call return- Returning Complex Objects, Functions are Objective, No Pointers in R, Recursion

(Bloom Level -BL4- BL5)

Assessment Tool Weightage: 25%

Code of Conduct:

I Adhithyan S (192421102) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

QUESTIONS: **Total Marks: 100**

Scenario

A retail company operates stores across multiple cities. The management has collected customer transaction data containing:

- Customer_ID
- Gender
- Age
- City
- Product_Category
- Purchase_Amount
- Monthly_Income
- Customer_Rating
- Membership_Status
- Store_Type

The company wants to improve customer satisfaction, optimize marketing strategies, and increase sales using data-driven decisions.

Question 1

The management wants to understand the overall spending behavior of customers. Calculate and interpret appropriate measures of central tendency (mean, median, and mode) for Purchase Amount. Based on the results, recommend which measure best represents customer spending patterns.

Question 2

Analyze the variability in customer spending using measures of dispersion such as range, variance, standard deviation, and interquartile range. Identify whether spending behavior is consistent across customers and justify your findings.

Question 3

The marketing team suspects that customer ratings differ significantly between premium and regular members. Perform an exploratory analysis of Customer Rating and summarize the key patterns using suitable statistical measures and visualizations.

Question 4

Investigate the relationship between Monthly Income and Purchase Amount using bivariate analysis techniques. Interpret the strength and direction of the relationship and discuss its implications for targeted marketing campaigns.

Question 5

The company wants to determine whether male and female customers differ in their average purchase amount. Formulate appropriate hypotheses, perform a T-test, and interpret the results to support business decision-making.

Question 6

Management wishes to compare customer satisfaction ratings across three different store types (Urban, Semi-Urban, and Rural). Conduct an ANOVA test and explain whether significant differences exist among the store categories.

Question 7

Analyze whether Membership Status and Product Category preference are associated. Perform a Chi-square test, interpret the findings, and explain how the results can support customer retention strategies.

Question 8

The company plans to launch a new loyalty program. Using EDA techniques, identify the customer segment that should be targeted first. Support your recommendation using appropriate statistical evidence.

Question 9

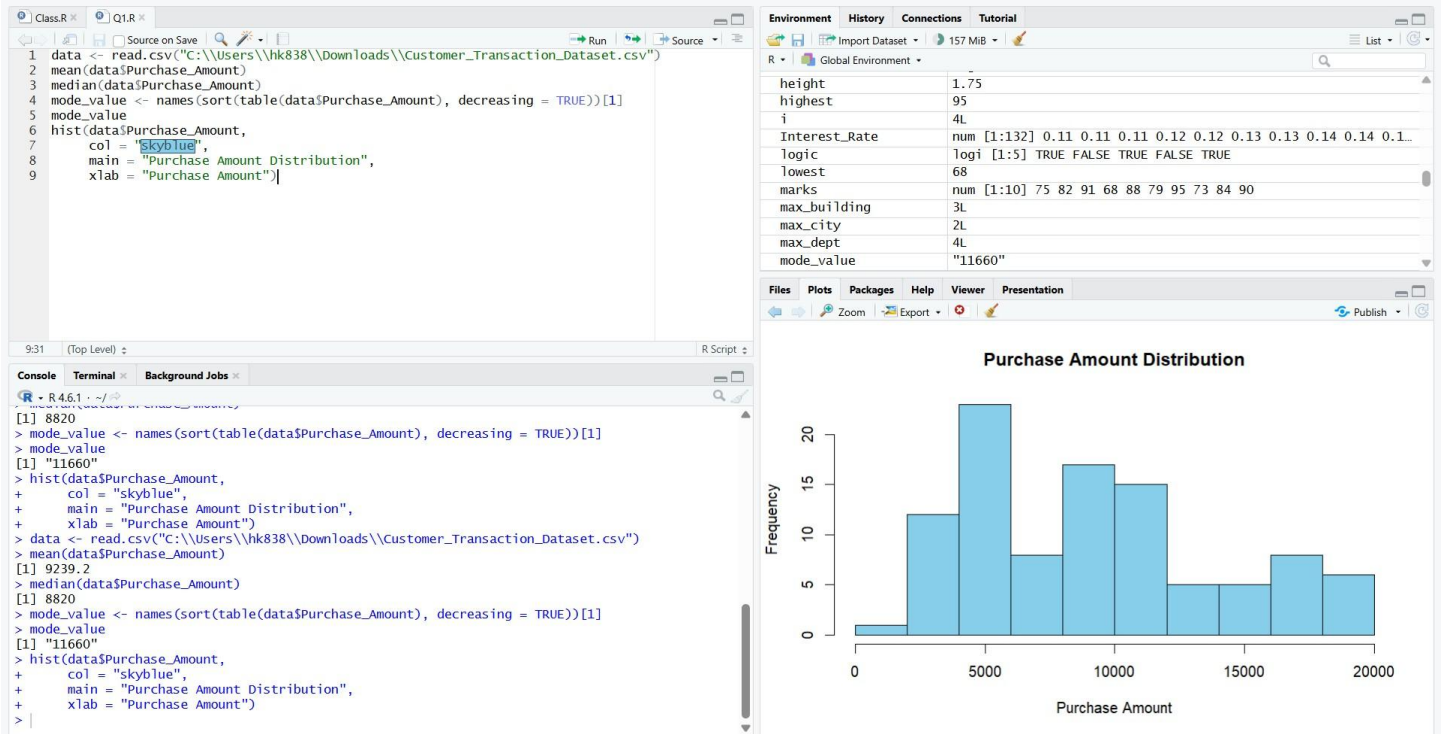
Based on customer transaction data, identify key factors influencing high purchase amounts. Utilize central tendency measures, dispersion measures, and bivariate analysis to justify your conclusions.

Question 10

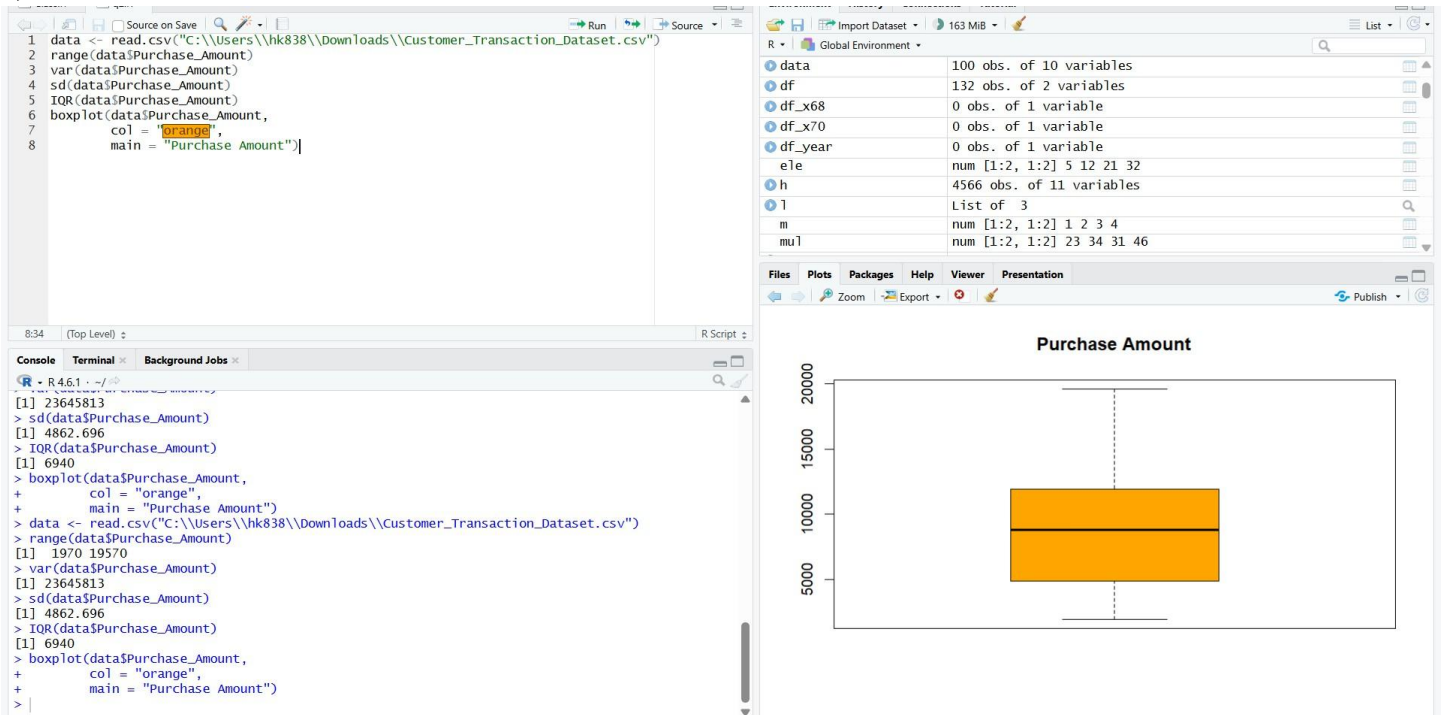
Prepare a business analytics report summarizing the insights obtained from EDA, T-test, ANOVA, and Chi-square analysis. Provide actionable recommendations that can help management improve customer engagement and profitability.

Answers:

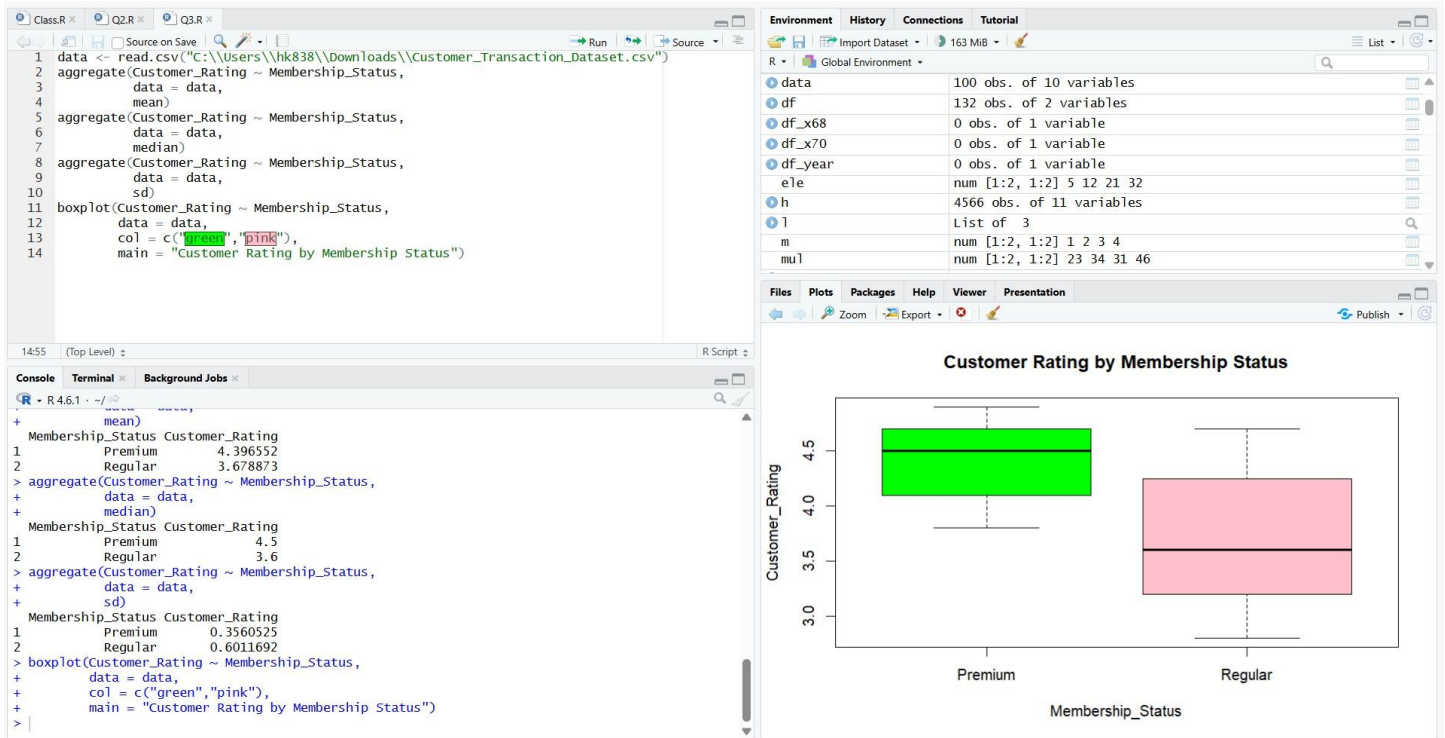
1)



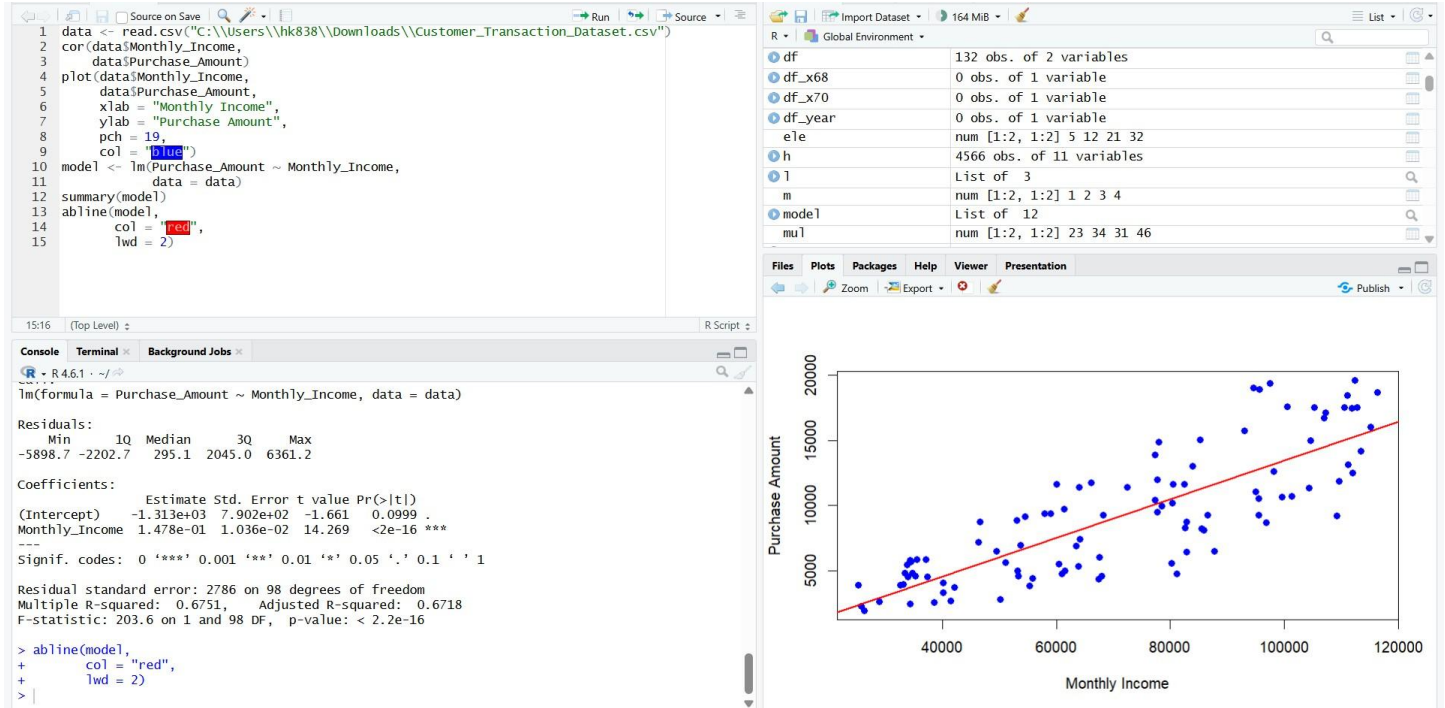
2)



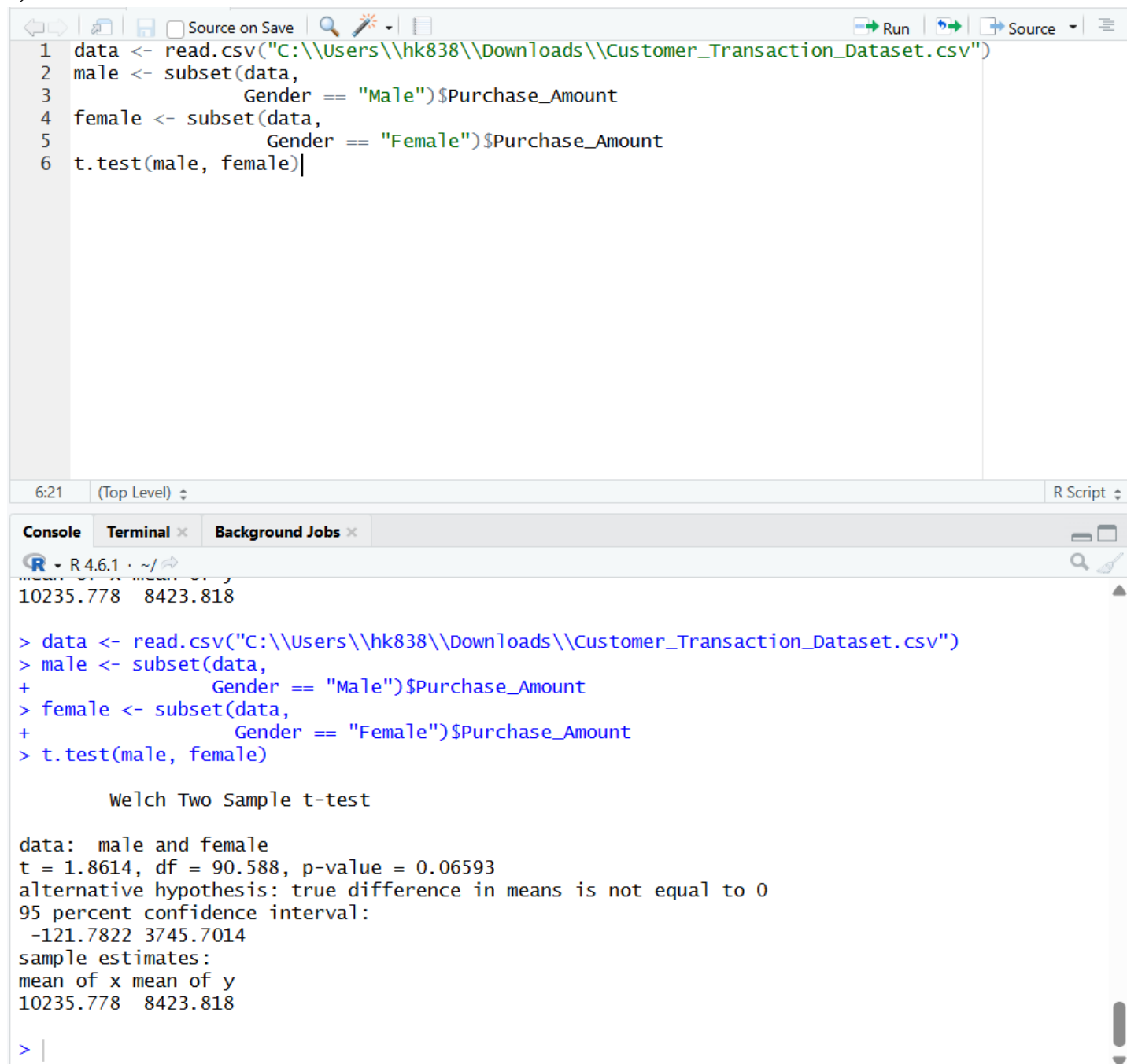
3)



4)



5)



The screenshot displays the R Studio environment. The top pane shows an R script with the following code:

```
1 data <- read.csv("C:\\Users\\hk838\\Downloads\\Customer_Transaction_Dataset.csv")
2 male <- subset(data,
3               Gender == "Male")$Purchase_Amount
4 female <- subset(data,
5                 Gender == "Female")$Purchase_Amount
6 t.test(male, female)|
```

The bottom pane shows the console output, which includes the results of the t-test:

```
R 4.6.1 ~ /
10235.778 8423.818

> data <- read.csv("C:\\Users\\hk838\\Downloads\\Customer_Transaction_Dataset.csv")
> male <- subset(data,
+               Gender == "Male")$Purchase_Amount
> female <- subset(data,
+                 Gender == "Female")$Purchase_Amount
> t.test(male, female)

Welch Two Sample t-test

data: male and female
t = 1.8614, df = 90.588, p-value = 0.06593
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
-121.7822 3745.7014
sample estimates:
mean of x mean of y
10235.778 8423.818

> |
```

6)

```

1 data <- read.csv("C:\\Users\\hk838\\Downloads\\Customer_Transaction_Dataset.csv")
2 anova_model <- aov(Customer_Rating ~ Store_Type,
3                     data = data)
4 summary(anova_model)
5 boxplot(Customer_Rating ~ Store_Type,
6          data = data,
7          col = c("yellow", "cyan", "pink"))

```

741 (Top Level) R Script

Console Terminal Background Jobs

```

R - R 4.6.1 ~ ./
> data <- read.csv("C:\\Users\\hk838\\Downloads\\Customer_Transaction_Dataset.csv")
> anova_model <- aov(Customer_Rating ~ Store_Type,
+                     data = data)
> summary(anova_model)
Df Sum Sq Mean Sq F value Pr(>F)
Store_Type  2  0.05  0.0242  0.06  0.942
Residuals 97 39.40  0.4062
> boxplot(Customer_Rating ~ Store_Type,
+          data = data,
+          col = c("yellow", "cyan", "pink"))
>

```



7)

```

1 data <- read.csv("C:\\Users\\hk838\\Downloads\\Customer_Transaction_Dataset.csv")
2 table_data <- table(data$Membership_Status,
3                     data$Product_Category)
4 table_data
5 chisq.test(table_data)
6 barplot(table_data,
7          beside = TRUE,
8          col = rainbow(5),
9          legend = TRUE)

```

826 (Top Level) R Script

Console Terminal Background Jobs

```

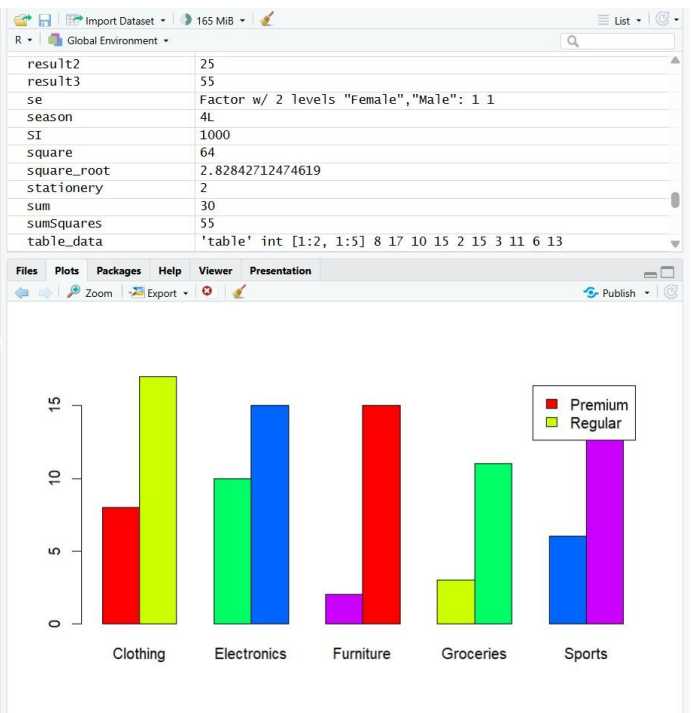
R - R 4.6.1 ~ ./
+ data$Product_Category)
> table_data
      Clothing Electronics Furniture Groceries Sports
Premium      8         10          2          3          6
Regular     17         15         15         11         13
> chisq.test(table_data)

Pearson's Chi-squared test

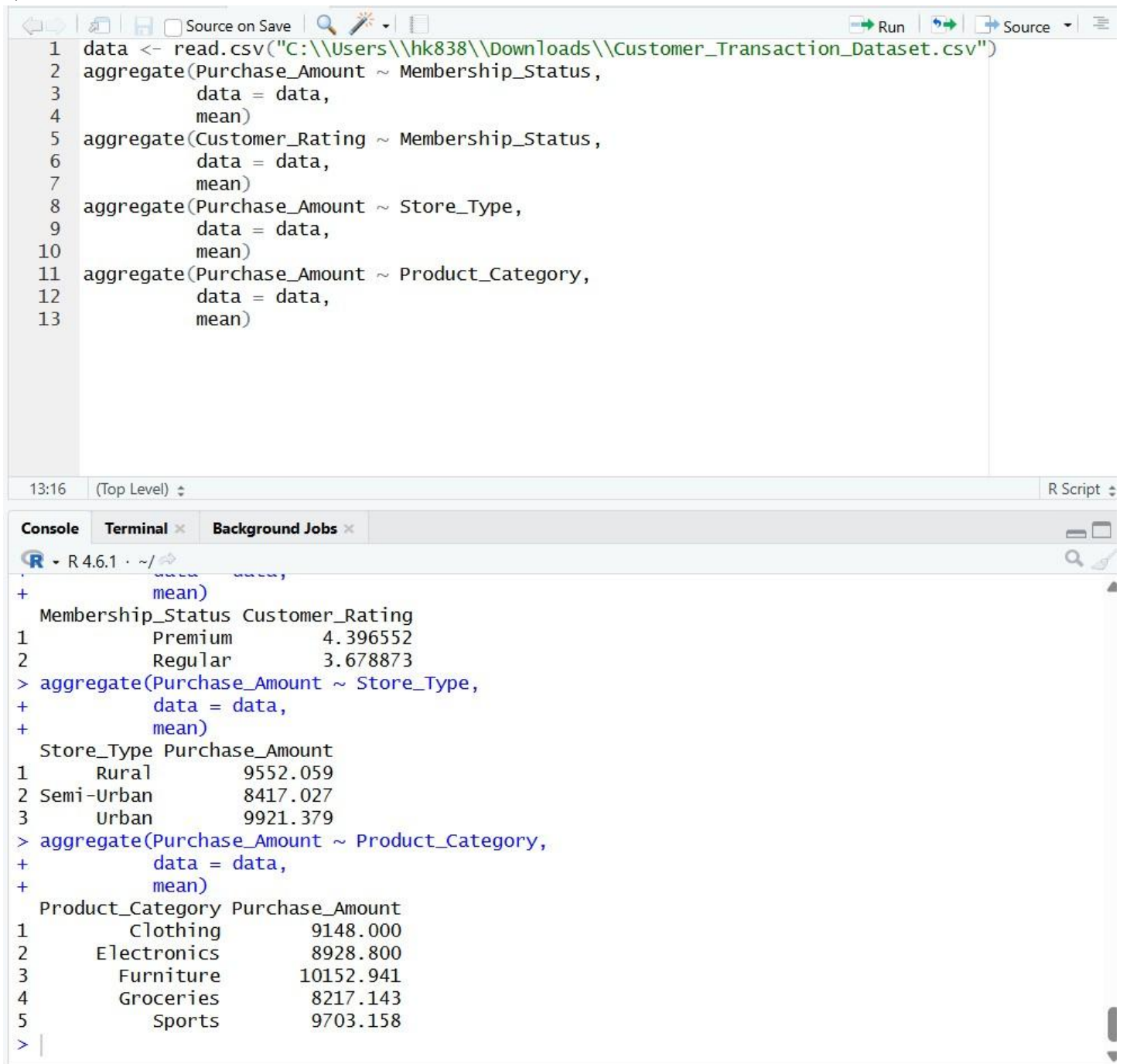
data: table_data
X-squared = 4.4822, df = 4, p-value = 0.3447

Warning message:
In chisq.test(table_data) : Chi-squared approximation may be incorrect
> barplot(table_data,
+          beside = TRUE,
+          col = rainbow(5),
+          legend = TRUE)
>

```



8)



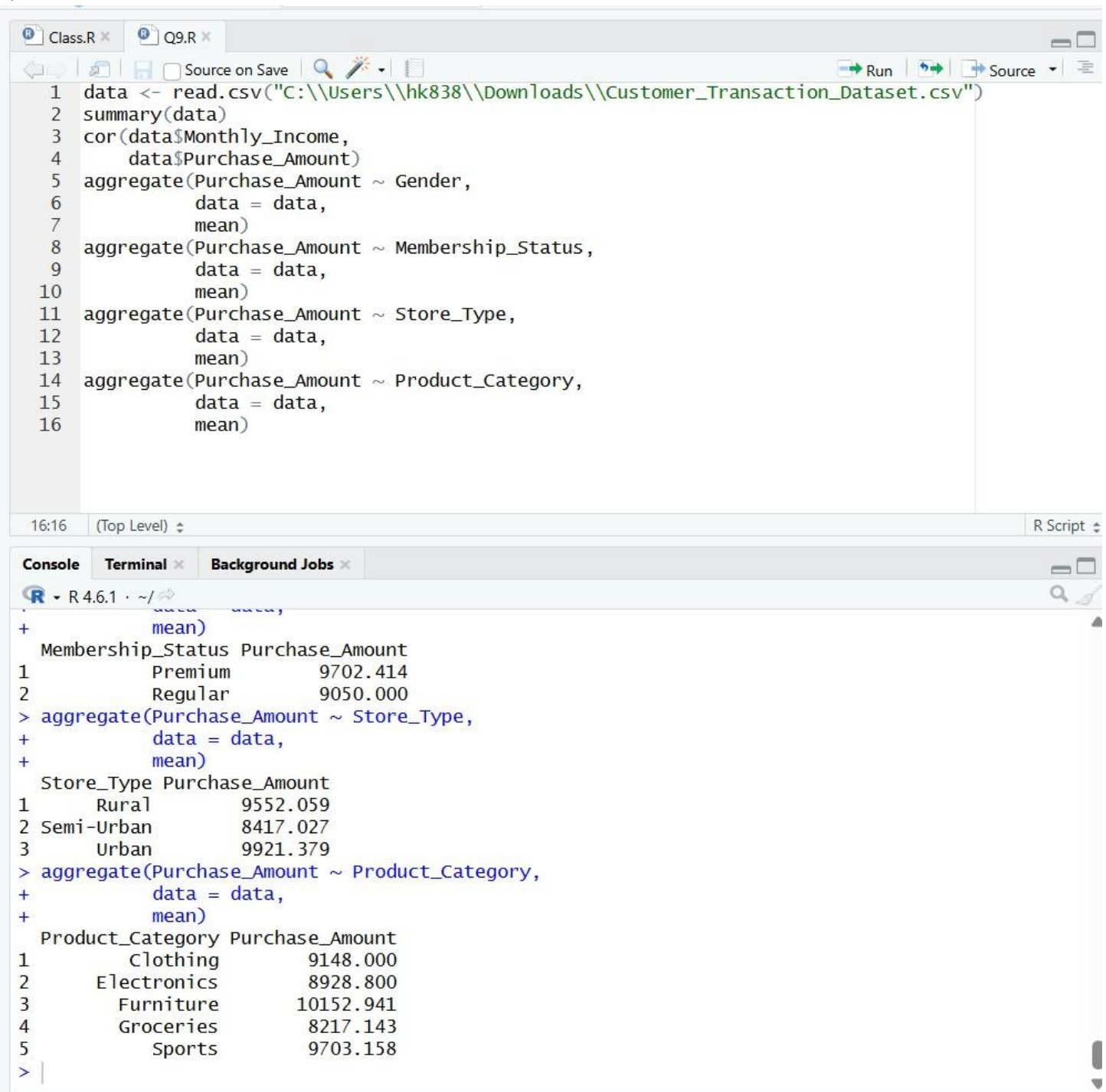
The screenshot displays the R Studio environment. The top pane shows the R script editor with the following code:

```
1 data <- read.csv("C:\\Users\\hk838\\Downloads\\Customer_Transaction_Dataset.csv")
2 aggregate(Purchase_Amount ~ Membership_Status,
3           data = data,
4           mean)
5 aggregate(Customer_Rating ~ Membership_Status,
6           data = data,
7           mean)
8 aggregate(Purchase_Amount ~ Store_Type,
9           data = data,
10          mean)
11 aggregate(Purchase_Amount ~ Product_Category,
12          data = data,
13          mean)
```

The bottom pane shows the console output, which includes the results of the first two aggregate functions and the start of the third:

```
R - R 4.6.1 - ~/...
+      mean)
+ Membership_Status Customer_Rating
1      Premium      4.396552
2      Regular      3.678873
> aggregate(Purchase_Amount ~ Store_Type,
+           data = data,
+           mean)
+ Store_Type Purchase_Amount
1      Rural      9552.059
2 Semi-Urban      8417.027
3      Urban      9921.379
> aggregate(Purchase_Amount ~ Product_Category,
+           data = data,
+           mean)
+ Product_Category Purchase_Amount
1      Clothing      9148.000
2      Electronics      8928.800
3      Furniture      10152.941
4      Groceries      8217.143
5      Sports      9703.158
> |
```


9)



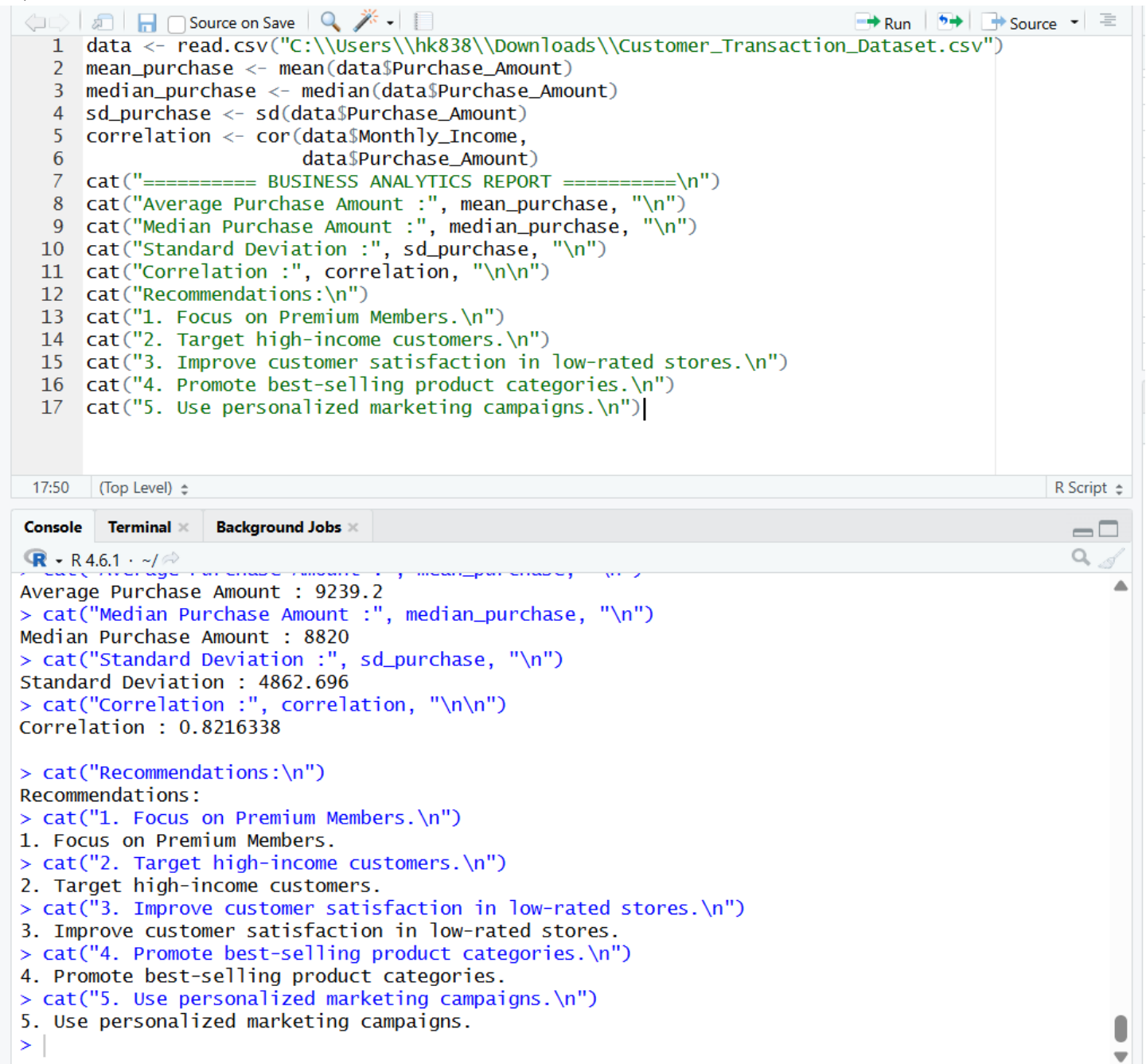
The screenshot displays the R Studio environment. The top pane shows a script with 16 lines of R code. The bottom pane shows the console output, which includes the results of the `aggregate` function for `Membership_Status`, `Store_Type`, and `Product_Category`.

```
1 data <- read.csv("C:\\Users\\hk838\\Downloads\\Customer_Transaction_Dataset.csv")
2 summary(data)
3 cor(data$Monthly_Income,
4     data$Purchase_Amount)
5 aggregate(Purchase_Amount ~ Gender,
6     data = data,
7     mean)
8 aggregate(Purchase_Amount ~ Membership_Status,
9     data = data,
10    mean)
11 aggregate(Purchase_Amount ~ Store_Type,
12    data = data,
13    mean)
14 aggregate(Purchase_Amount ~ Product_Category,
15    data = data,
16    mean)
```

Console Output:

```
R > R 4.6.1 ~ /
+      mean)
+ Membership_Status Purchase_Amount
1      Premium      9702.414
2      Regular      9050.000
> aggregate(Purchase_Amount ~ Store_Type,
+     data = data,
+     mean)
+ Store_Type Purchase_Amount
1      Rural      9552.059
2 Semi-Urban      8417.027
3      Urban      9921.379
> aggregate(Purchase_Amount ~ Product_Category,
+     data = data,
+     mean)
+ Product_Category Purchase_Amount
1      Clothing      9148.000
2      Electronics  8928.800
3      Furniture  10152.941
4      Groceries   8217.143
5      Sports     9703.158
> |
```

10)



The screenshot displays the R Studio environment. The top pane shows an R script with 17 lines of code. The code reads a CSV file, calculates statistical measures (mean, median, standard deviation, correlation), and prints a business analytics report. The bottom pane shows the console output, which matches the printed statements in the script.

```
1 data <- read.csv("C:\\Users\\hk838\\Downloads\\Customer_Transaction_Dataset.csv")
2 mean_purchase <- mean(data$Purchase_Amount)
3 median_purchase <- median(data$Purchase_Amount)
4 sd_purchase <- sd(data$Purchase_Amount)
5 correlation <- cor(data$Monthly_Income,
6                   data$Purchase_Amount)
7 cat("===== BUSINESS ANALYTICS REPORT =====\n")
8 cat("Average Purchase Amount :", mean_purchase, "\n")
9 cat("Median Purchase Amount :", median_purchase, "\n")
10 cat("Standard Deviation :", sd_purchase, "\n")
11 cat("Correlation :", correlation, "\n\n")
12 cat("Recommendations:\n")
13 cat("1. Focus on Premium Members.\n")
14 cat("2. Target high-income customers.\n")
15 cat("3. Improve customer satisfaction in low-rated stores.\n")
16 cat("4. Promote best-selling product categories.\n")
17 cat("5. Use personalized marketing campaigns.\n")
```

Console Output:

```
R 4.6.1 : ~/
> cat("Average Purchase Amount :", mean_purchase, "\n")
Average Purchase Amount : 9239.2
> cat("Median Purchase Amount :", median_purchase, "\n")
Median Purchase Amount : 8820
> cat("Standard Deviation :", sd_purchase, "\n")
Standard Deviation : 4862.696
> cat("Correlation :", correlation, "\n\n")
Correlation : 0.8216338

> cat("Recommendations:\n")
Recommendations:
> cat("1. Focus on Premium Members.\n")
1. Focus on Premium Members.
> cat("2. Target high-income customers.\n")
2. Target high-income customers.
> cat("3. Improve customer satisfaction in low-rated stores.\n")
3. Improve customer satisfaction in low-rated stores.
> cat("4. Promote best-selling product categories.\n")
4. Promote best-selling product categories.
> cat("5. Use personalized marketing campaigns.\n")
5. Use personalized marketing campaigns.
> |
```

RUBRICS FOR CALCULATION

Industry Problem Solving Task					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Application of EDA Techniques	25	Correctly applies central tendency, dispersion, and bivariate analysis techniques.	Applies most EDA techniques correctly with minor omissions.	Applies basic EDA techniques with limited accuracy.	Demonstrates limited or incorrect application of EDA techniques.
Statistical Analysis & Testing	25	Correctly performs and interprets T-test, ANOVA, and Chi-square analysis.	Performs statistical tests with minor interpretation errors.	Performs basic statistical analysis with limited interpretation.	Demonstrates poor understanding of statistical tests.
Problem Analysis & Data Interpretation	20	Accurately analyzes data patterns and derives meaningful insights.	Provides reasonable analysis with adequate interpretation.	Provides limited analysis and interpretation.	Analysis is weak or incorrect.
Decision-Making & Business Recommendations	20	Provides data-driven recommendations that effectively address the business problem.	Recommendations are relevant and supported by analysis.	Recommendations are basic with limited justification.	Recommendations are weak, unsupported, or missing.
Organization & Completeness	10	Response is well-structured, complete, and addresses all requirements.	Mostly complete and organized.	Partially complete with some missing components.	Incomplete and poorly organized.